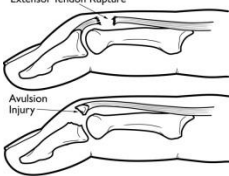


EMD Backslab / Splint Guide (Oct 2017)

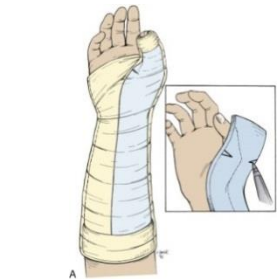
*this is just a quick reference guide and was created on a presumption that you know how to make a back-slab. Do consult your P3 seniors if you have doubts about anything at all!

*for closed #s only - All open # warrants hand/ortho review amongst other things

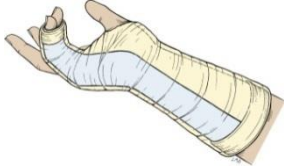
*Process owner: Dr Tan Chieh Ling Jacqueline

Fracture or Dislocation Site	Management Backslab /Splint site and length	Angles	Diagram
Fingers excluding thumb			
Fractures			
Distal phalanx #	Buddy splint Short-term immobilization with a dorsal or volar splint can be used for added protection or pain control	PIPJ and DIPJ in extension (not hyperextension)	
Middle phalanx #			
Prox phalanx #			
Mallet finger 	Zimmer splint – volar or dorsal aspect, extending from tip of finger to just prox to the PIPJ (to allow some movement at PIPJ)	DIPJ in extension (not hyperext)	
Dislocations			
DIPJ dislocation	<u>M&R under digital block</u> Place Zimmer splint over dorsal aspect after M&R	DIPJ in 20-30deg flexion	
PIPJ dislocation – dorsal	<u>M&R under digital block</u> Buddy splint after M&R	PIPJ and DIPJ in extension	
PIPJ volar-avulsion # with dorsal dislocation 	<u>M&R under digital block</u> Place extension block splint over dorsal aspect after M&R	MCPJ – 70-80deg flexion PIPJ – 45 deg flexion Secured at the proximal finger, allowing flexion at the PIP joint	

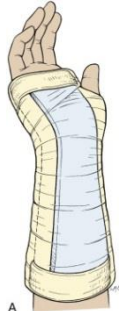
Common ligamentous/soft tissue injuries

Volar plate sprain Tenderness over PIPJ/DIPJ following hyperextension injury	Mild hyperextension injury – Buddy splint Avulsion fractures or suspected volar plate disruptions - splinted with an extension block splint	PIPJ and DIPJ – full extension PIPJ – 45 deg flexion	
Collateral ligament sprain (excluding thumb)	Buddy splint		
Central slip rupture: Loss of active extension over the PIPJ	Volar splint – across middle and proximal phalanx	PIPJ – full extension	
Flexor tendon injury of DIPJ Inability to actively flex DIPJ following rupture of the FDP insertion	Dorsal splint extending from tip of finger to base of prox phalanx	DIPJ and PIPJ in slight flexion	
Thumb and 1st MC			
Distal and proximal phalangeal # of the thumb	Thumb spica – depending on location of the #, to include distal joint up till proximal forearm (at least 2FB distal to elbow)	Wrist at 0-15deg of dorsal extension Thumb in opposition to index finger	
Base of 1st MC fracture (regardless of type) 	Thumb spica – including 1st MCPJ till proximal forearm (at least 2FB distal to elbow) Leave thumb IPJ free	Wrist at 0-15deg of dorsal extension Thumb in opposition to index finger	

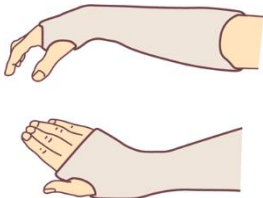
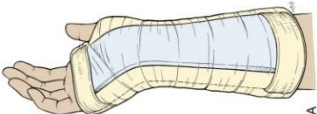
Metacarpals

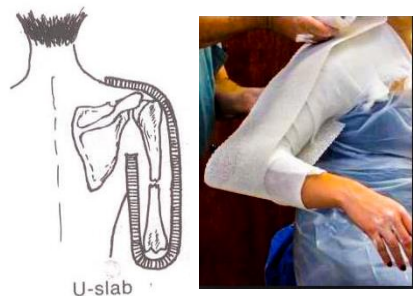
2 nd and 3 rd MC #	Radial gutter splint – distal to DIPJ of 2 nd and 3 rd MC till proximal forearm (at least 2FB distal to elbow) over radial side	Forearm – neutral Wrist – 0-15 deg extension MCPJ - 50 deg flexion PIPJ/DIPJ - 5 to 10 deg flexion	
4 th and 5 th MC #	Ulnar gutter splint – distal to DIPJ of 4 th and 5 th MC till proximal forearm (at least 2FB distal to elbow) over ulnar side	Forearm – neutral Wrist – 0-15 deg extension MCPJ - 50 deg flexion PIPJ/DIPJ - 5 to 10 deg flexion	

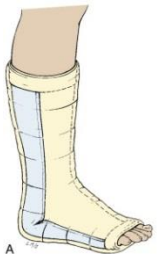
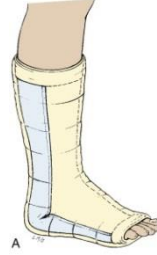
Carpal bones

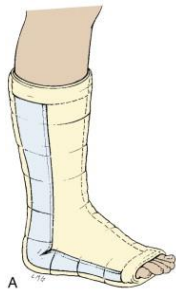
Scaphoid #	Thumb spica – including 1 st MCPJ till proximal forearm (at least 2FB distal to elbow) Leave thumb IPJ free	Wrist at 0-15deg of dorsal extension Thumb in opposition to index finger	
Trapezium #			
All other carpal bone #	Volar short arm slab – MC to proximal forearm (at least 2FB distal to elbow)	Wrist - 30-50 deg extension	
<u>Lunate and perilunate dislocations</u>	<u>Consult HRM</u>		

Radius-Ulna

Colles Fracture (distal radius extra-articular)	Dorsal short arm slab – MC to proximal forearm (at least 2FB distal to elbow)	Forearm – pronation Wrist – slight flexion	 Source: John Murtagh, Jill Rosenblatt: John Murtagh's General Practice, 6e: www.murtagh.mhmedical.com Copyright © McGraw-Hill Education. All rights reserved.
Smith's Fracture (distal radius extra-articular)	Volar short arm slab – MC to proximal forearm (at least 2FB distal to elbow) Add on a long arm backslab with elbow at 90deg (as fracture is unstable)	Forearm – supination Wrist – dorsiflexion	

Shaft of radius and ulna	For Ortho review, pt will most likely need to be admitted, meanwhile can immob with a long arm backslab or sugar tong backslab		
Monteggia and Galeazzi			
Radial head #	Undisplaced - Long arm backslab – Start on the posterior aspect of the proximal arm, down the ulnar aspect of arm, ends at the MCP joints Displaced - Refer Ortho	Elbow - 90 deg Forearm - neutral position with thumb up Wrist - neutral or slightly extended	
Humerus			
Shaft of humerus	<u>Low velocity</u>, min displaced, close # - U slab – extending from the axilla, around the elbow to above the gleno-humeral joint <u>High velocity polytrauma, open #, severe soft tissue injury or bone loss</u> - <u>Refer ortho, KIV admission for ORIF</u>	Elbow joint at 90deg	
Surgical neck of humerus	Collar and cuff		

Fracture or Dislocation Site	Management Backslab /Splint site and length	Angles	Diagram
Phalangeal and metatarsal injuries/fractures			
Fractures			
Single distal, middle or proximal phalangeal #	Buddy splint/tap		
Multiple toes phalangeal #	Buddy splint/tap verses below knee backslab	Ankle in 90deg dorsiflexion, without inversion/eversion	
Metatarsal #	Below knee backslab	Ankle in 90deg dorsiflexion, without inversion/eversion	
Lisfranc injury (Tarsometatarsal fracture-dislocation)	For Ortho review <i>*missed injuries can result in progressive foot deformity, chronic pain and dysfunction</i>		
Ankle injuries			
Ankle sprain	Jones Bandage Consider below knee backslab <i>Refer to ankle sprain workflow for management</i>	Ankle in 90deg dorsiflexion, without inversion/eversion	

Ankle fracture (undisplaced or minimally displaced lat or med malleolus fracture, without intra-articular involvement)	Below knee backslab <i>Consider XR of the knee to look for/rule out maisonneuve injury</i>	Ankle in 90deg dorsiflexion, without inversion/eversion	
Ankle fracture (bi-malleolar, unstable)	For Ortho review		
Ankle dislocation	M&R with Entonox For XR and Ortho review after		
Achilles Tendon Rupture	For Ortho review if complete rupture Consider above knee backslab if it is a partial tear	Ankle in plantarflexion, knee in slight (10-15deg) flexion	
Tib/Fib #			
Proximal or shaft of tib/fib fractures	For Ortho review – mostly due to high energy trauma, may be open fractures as little soft tissue cover If for conservative management, place an above knee backslab	Ankle in 90deg dorsiflexion, without inversion/eversion, knee in slight (10-15deg) flexion	
Patella/knee injuries			
Patella dislocation	Above knee backslab	Ankle in 90deg dorsiflexion, without inversion/eversion, knee in slight (10-15deg) flexion	
Patella fracture with intact extensor mechanism	Above knee backslab		
Patella fracture with extensor mechanism failure	For Ortho review – need surgical repair		
Stable soft tissue injuries of the knee with swelling	Jones bandage		

References

https://www.cfps.org.sg/publications/the-singapore-family-physician/article/940_pdf

Guide to Essentials in Emergency Medicine – Edited by Prof Shirley Ooi and Prof Peter Manning

<http://www.orthobullets.com>